Tilak Raghunandan G

AI & Machine Learning Engineer · **Computer Vision** · **NLP & LLMs** · **Full-Stack Developer**

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PROFESSIONAL SUMMARY

Results-driven AI & Machine Learning undergraduate with hands-on internship experience engineering **RAG pipelines** and **LLM-backed systems** at a technology company, and a strong competitive track record with **3 podium finishes** at state- and national-level hackathons. I build things that work — from a **95% accuracy medical diagnostics system** to a real-time multilingual emergency platform. Equally comfortable with deep learning research and full-stack product delivery. Looking for roles in **AI/ML Engineering**, **Computer Vision**, or **NLP** where I can contribute from day one.

EXPERIENCE

AI / ML Engineering Intern — *Oct 2025 – Feb 2026*
Unriddle Technologies — Shivamogga, Karnataka

- Designed and deployed end-to-end **Retrieval-Augmented Generation (RAG) pipelines** powered by **LLMs**, **vector embeddings**, and **semantic search** over large document corpora, cutting retrieval latency and significantly improving answer relevance.
- Applied **prompt engineering** strategies and fine-tuned retrieval parameters to boost Q&A accuracy; validated improvements through automated evaluation benchmarks.
- Built and maintained **NLP preprocessing** and **text chunking** workflows for multi-document knowledge retrieval, collaborating in an **Agile** team environment to ship production-adjacent ML systems.
- Integrated **embedding generation** pipelines and **semantic similarity** scoring into a document intelligence product, improving output coherence across diverse query types.

Volunteer Coordinator — *Sep 2024 – Present*
Anveshana Startup Summit, PES Institute of Technology and Management

- Led end-to-end operations for a large-scale multi-day innovation summit, coordinating cross-functional teams and enabling high-value networking between 50+ startups and industry mentors.

HACKATHONS & ACHIEVEMENTS

1st Runner-Up — *March 2026*
Presidency University Hackathon (48-hour sprint)

- Built and presented an award-winning AI solution tackling a complex real-world problem, beating 100+ competing teams.

2nd Runner-Up — *Oct 2025*
DIGITISE 2K25 — Adichunchanagiri Institute of Technology, Chikkamagaluru

- Architected a real-time **computer vision pipeline** for automated agricultural quality control, earning podium placement among 80+ teams.

Top 5 Finalist — *May 2025*
Bapuji Institute of Technology Hackathon

- Recognized for the Multilingual Emergency Support Platform — a real-time NLP + speech translation system bridging language gaps between rural doctors and patients.

Participant (National Level) — *2025*
MONAITHON 2025, JNNCE Shivamogga · Agentic AI Day (Google Cloud × Hack2Skill) · GDG Solution Challenge 2025

PROJECTS

AI Retinal Disease Detection System — *2025*
Academic Research Project — Python · PyTorch · CNN · U-Net · ResNet · VGG16 · Transfer Learning · OpenCV

- Built a multi-class retinal disease classifier using **transfer learning** on ResNet and VGG16, with a **U-Net** segmentation head for blood vessel mapping on labelled ophthalmology datasets.
- Applied **data augmentation** and class-balancing to handle imbalanced medical data; implemented full **model evaluation** including ROC-AUC, confusion matrices, and per-class F1 scoring.
- Achieved **95% classification accuracy** — benchmarked against baseline CNNs, demonstrating clinical-grade reliability suitable for diagnostic support tooling.

Multilingual Emergency Support Platform — *2025*
Hackathon Project — Top 5 Finalist — React.js · Node.js · NLP · LLM APIs · Speech-to-Text · AI Translation

- Developed a full-stack rural healthcare platform integrating **real-time speech-to-text**, **NLP-powered mul-**

tilingual translation, and **LLM-based medical triage** to remove language barriers between doctors and patients.

- Designed for low-connectivity rural environments with a responsive React.js frontend and Node.js REST API backend — recognised as **Top 5 Finalist** at Bapuji Institute of Technology.

Vegetable Quality Classification System *2025*
Academic Project — 2nd Runner-Up, DIGITISE 2K25 Python · OpenCV · CNN · Scikit-learn · Real-Time Inference

- Engineered a **real-time CV inference pipeline** using image segmentation and ML classifiers to automate agricultural produce sorting, replacing manual inspection at processing facilities.
- Optimised for edge deployment with fast inference and high classification accuracy across 15+ vegetable categories.

Fitness & Calorie Tracker — Full-Stack Web App *2024*
Personal Project React.js · Node.js · MySQL · REST API · Responsive Design

- Built a production-ready full-stack web application with a **RESTful API** (Node.js / MySQL) and a React.js dashboard for meal logging, calorie goal tracking, and fitness analytics.
- Designed a normalised relational schema and modular API endpoints for secure, reliable data persistence.

TECHNICAL SKILLS

ML / Deep Learning	PyTorch, Scikit-learn, CNNs, Transfer Learning, U-Net, ResNet, VGG16, Data Augmentation, Model Evaluation, Hyperparameter Tuning, Feature Engineering
NLP & LLMs	RAG Pipelines, LLM Integration, Prompt Engineering, Embeddings, Semantic Search, Vector Databases, Speech-to-Text, AI Translation, NLP Preprocessing, Hugging Face (Transformers)
Computer Vision	OpenCV, Image Segmentation, Object Detection, Real-Time Inference, Image Processing, Feature Extraction, Medical Image Analysis
Web & Full-Stack	React.js, Node.js, HTML5, CSS3, REST APIs, PHP, Responsive Design
Databases	MySQL, MongoDB, Vector Databases, Relational Schema Design
Programming	Python (primary), C, Java (basic), Bash (basic)
Tools & Platforms	Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Linux CLI, Google Cloud (GCP), n8n, Postman
Soft Skills	Problem Solving, Cross-functional Collaboration, Agile Teamwork, Technical Communication, Adaptability
Languages	English (Fluent) | Hindi (Proficient) | Kannada (Native)

EDUCATION

Bachelor of Engineering (B.E.) — Artificial Intelligence & Machine Learning *Jan 2023 – Jan 2027 (Expected)*
PES Institute of Technology and Management, Shivamogga **CGPA: 7.35 / 10**

- Relevant coursework: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Database Management Systems, Operating Systems, Linear Algebra & Statistics
- Active participant in national hackathons, applied research projects, and campus innovation events

Pre-University Education — PCMCs / Science *Jan 2022 – Jan 2023*
DVS Composite PU College, Shivamogga